

# ClaimLens — API Documentation

Base URL: <http://127.0.0.1:8000/api/v1> | Interactive OpenAPI Docs: <http://127.0.0.1:8000/docs>

## 1. Health Check Endpoint

**GET /health** — Returns system readiness, vector index count, and model initialization status.

**Sample Response:**

```
{"status": "healthy", "service": "ClaimLens - AI-Powered Insurance Claims Assistant",  
"vector_store_indexed_count": 500, "ml_models_ready": true}
```

## 2. Input Validation Guardrails

**POST /claims/validate** — Validates claim schema, range constraints, and sanitizes prompt injections.

**Sample Request:**

```
{"claim_type": "Auto", "state": "CA", "policyholder_tenure_years": 4.5, "previous_claims_count": 0,  
"incident_date": "2024-01-10", "claim_filed_date": "2024-01-14", "claim_amount": 3500.0, "deductible":  
500.0}
```

**Sample Response:**

```
{"is_valid": true, "errors": [], "warnings": [], "sanitized_data": {...}}
```

## 3. Hybrid RAG Search Endpoint

**POST /claims/hybrid-search** — Retrieves similar historical claims using in-memory vector search + BM25 keyword matching with RRF scoring.

**Sample Request:**

```
{"query": "rear fender collision repair", "top_k": 3, "claim_type": "Auto", "state": "CA"}
```

**Sample Response:**

```
{"query": "rear fender collision repair", "count": 3, "results": [{"claim_id": "CLM-0000000001",  
"hybrid_score": 0.9412, "retrieval_method": "Hybrid", "claim_amount": 3200.0, "claim_status": "Approved"}]}
```

## 4. Classification & Prioritization Endpoint

**POST /claims/classify** — Classifies claim complexity and assigns SLA priority triage score.

**Sample Response:**

```
{"complexity_level": "Low / Standard Complexity", "priority_tier": "P4 - Low / Routine", "priority_score": 25.0, "target_sla_resolution_hours": 120}
```

## 5. Predictive Machine Learning Endpoint

**POST /claims/predict** — Predicts claim payout amount, resolution days, and fraud probability with explainability.

**Sample Response:**

```
{"predicted_claim_amount": 3680.0, "estimated_amount_range": "$3,128.00 - $4,416.00", "predicted_days_to_resolution": 14.2, "fraud_probability_percent": 18.5, "risk_tier": "Low Risk", "top_risk_drivers": ["Claim attributes match normal underwriting patterns."]}
```

## 6. Multi-Agent End-to-End Analysis Workflow

**POST /claims/analyze** — Executes the complete 3-agent orchestration (Investigation → Risk → Recommendation) and LLM-as-Judge audit.

**Sample Response:**

```
{"classification": {...}, "ml_predictions": {...}, "investigation": {"agent_name": "Investigation Agent", "findings": "..."}, "risk_assessment": {"fraud_probability_percent": 18.5}, "recommendation": {"decision": "AUTO_APPROVE", "fast_track_eligible": true, "recommended_payout": "$3,000.00"}, "llm_as_judge_review": {"verdict": "PASS", "overall_quality_score": 9.3}}
```

## 7. Adjuster Feedback Loop Endpoints

**POST /claims/feedback** — Records human adjuster reviews (Approve/Modify/SIU) into persistent store.

**GET /claims/feedback/metrics** — Returns total reviews and AI agreement rate analytics.